Occultation rate

SLANT network study. Computed 8 October 2026 for 100 host orbits over one year.

The slit plate is fixed to the host and faces backward along the orbit (the wake), so it records orbital sunsets. This note counts how many occultations one unit records when the plate's field of view is applied, and what that rate sets for data volume and coincidences between units.

Field of view

The plate accepts sunlight arriving within 40.0° of its normal in elevation and 51.4° cross-track. One occultation sweeps the Sun through ±15.3° of elevation about the centre of the sweep, which leaves ±24.7° for mounting error. At sunset the Sun lies behind the spacecraft and at sunrise ahead of it, so a plate facing the wake records sunsets only.

Method

The host population is 100 orbits, propagated with J2 from 1 January 2026 for 360 days with a low-precision solar ephemeris. For each event the sweep is sampled every 5 s, from the reference frame (Sun 10° above the local horizontal) to the deep-limb frame (tangent altitude 60 km). The plate normal lies in the orbit plane along the wake direction, tilted to the centre of the sweep for that host's altitude. An event counts when every sample has the Sun in front of the plate within 40.0° in elevation and 51.4° cross-track. Mounting error is taken as zero.

Results

Quantity	Every 200 km crossing, no field of view	Wake-facing, in the field of view
Occultations per unit per day, mean over a year	27.0 (range 7.5–30.9)	**10.3** (range 0–15.4)
Unit-days with at least one occultation	91 %	**70 %**
Per day, 6 / 30 / 100 units	170 / 851 / 2,699	**68 / 329 / 1,033**
One day, 1 January 2026, 6 / 30 / 100 units	153 / 636 / 2,459	**47 / 189 / 940**
Coincident pairs per year within 500 km and 30 min, 100 units	1.32 million	**0.352 million**
Same, 6 / 30 units	5,400 / not computed	**1,558 / 47,804**
Largest beta angle for an occultation in view	66–69°	**about 48°** (accepted maximum 48.8°, rejected minimum 46.9°)

The first column counts every crossing of the 200 km tangent altitude, sunrises and sunsets alike, so the difference between the columns is the field of view alone.

Coincident pairs fall to 27 % of the unrestricted count while occultations fall to 38 %. Every wake-facing event is a sunset, and sunset events pair with each other: 0.38 *N* events form about (0.38

$N)^2$ pairs, 29 % of the $N^2/2$ pairs formed when dawn and dusk events were both counted, close to the 27 % found.

By host group the rate is 11.6 a day for the 53° shell, 10.9 for mid inclinations, 15.3 for low inclinations, 8.3 for near-polar orbits and 7.8 for sun-synchronous orbits. A plate facing the ram direction gives the same rate, because each orbit has one sunrise and one sunset at the same beta angle.

Local time. Tangent points of wake-facing events fall mostly at the evening terminator: 82 % between 15 and 24 h local solar time (40 hosts, 30 days). Ram-facing events fall mostly at dawn (60 % between 6 and 9 h).

Aperture

The 88 mm aperture, 27.72 mm above the slits, limits the two angles together. Every point of every slit clears the aperture at the full 40.0° of elevation for cross-track angles up to 44.0°; at the 51.4° cross-track limit the slit corners clear it only up to 22.9° of elevation. The centre of the slit field clears the aperture everywhere inside the detector limits.

Criterion at every frame of the sweep	Per unit per day
Detector limits only	10.33
Detector limits, and every point of every slit clears the aperture	10.14
Detector limits, and the centre of the slit field clears the aperture	10.33

In 1.85 % of events part of a slit is blocked in some frames. In those the worst slit loses a median 8.5 % of its length (99th percentile 28 %, maximum 33 %), over at most 8.5 % of the frames. These events are kept, so the rate is 10.3 a day. The retrieval excludes the blocked rows, which follow from the Sun vector in the instrument frame.

What the rate sets

With a pass of 467 s, 10.3 occultations a day give a duty cycle of 5.6 % and 7.2 MB of data a day (670 bit/s averaged). With CAN 2.0B framing, a day's data takes 2.0 to 2.4 minutes on a 1 Mbit/s bus.

A wake-facing network samples the evening terminator. Dawn coverage needs ram-facing units, whose filters face the atomic-oxygen flux; the network study on the Cost and scalability tab compares the two orientations.

Code: `channel-plan/net_fov.py`, with the aperture criteria in `net_fov_ap.py`, in the code archive.